MGTA 611
Deep Learning Project

Overview of the Deep Learning Project

- You and your group will develop a compelling business problem and acquire a dataset to solve (or make progress towards a solution) the problem using descriptive, visual and predictive (machine learning) analysis

Goals:

- Acquire hands on experience applying advanced analytic techniques to a business problem *you are interested in*
- Formulate convincing and persuasive reports and presentations
- Clearly and persuasively use the results of quantitative analysis to argue for a particular course of action
- Demonstrate how business data and analytical results can be used as an ongoing tool to further a company's ambitions

- Groups are self-selected and consist of 1-4 students each
- To 'declare' your group, create and share your project backlog (Excel file) with me
  - Template of the backlog file is available on MyLearningSpace

Project Components: Project Management

- Active project management is required throughout the project
- Agile Project Management
  - Maintain a project backlog
  - Biweekly sprints: It is expected that the backlog will be *meaningfully* updated at least every 2 weeks
- Project backlog (with group member's names) must be in place by January 19th – 2.5% of project grade
- Evaluation of backlog – 10% of project grade
  - Quality and consistency of backlog updates
  - May be used to individually adjust final grades

Product backlog

MGTA611	January - April 2026			
Lazaridis School of Business and Economics, Wilfrid Laurier University				
Project title	*title*			
Group name	*group name*			
Group members	Name	Student num	In progress	Completed tasks
	1 member 1	123456789	1	1
	2		0	0
	3		0	0

Project requirement	Task	Assigned to	Due date	Status	Completion date	Notes
Team declaration	*open product backlog file*	*member 1*	*9 Jan 2026*	*complete*	9 Jan 2026	
Team declaration	*share product backlog w. teaching team*	*member 1*	*?*	*in progress*		*project and group member information and share with the mrblair@wlu.ca via OneDrive*

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Project Components: Proposal

Expectations

- Acquired a dataset
- Chosen a business setting and problem
- Completed preliminary analysis – i.e. Requirements analysis, component design, and exploratory analysis of the dataset

Objective

- Convince the reader that: (a) there is an important problem to solve and (b) there is a feasible course of action available to solve this problem

Deliverable

- PDF report of at most six pages (excluding citations)

Evaluation

- 25% of project grade
- Professional and clear report that presents the problem and proposed solution in a compelling fashion to achieve the above objectives
- Report should include a detailed description of the analysis that will be carried out

Project Components: Final Report & Package

- The Final Report consists of three components
  1. A two-page executive summary of your project and results (excluding figures and citations)
  2. A Jupyter notebook which outlines and duplicates the key results in your descriptive and predictive analyses. It should show the development and improvement of the predictive model
  3. A PDF version of the Jupyter notebook
- Remember to appropriately city any resources uses including AI tools
  - You are responsible for, and are expected to understand, every sentence of the report and line of code in the Jupyter notebook
  - In exceptional cases you may be asked to discuss the documents with the professor
- Evaluation – 42.5% of project grade
  - Executive summary should be professional and clearly highlights the key pieces of the problem and the results
  - Jupyter notebook illustrates a successful implementation of the chosen analyses

Project Components: Presentation & Discussion

Requirements

- Oral presentation to the class (length determined by the number of groups, but likely 5-8 minutes)
- Respond to questions from the professor and your peers
- Groups will be assigned another group that they will pose questions to
- Q&A will be longer (e.g. 10-15 minutes)

Objectives

- In a short period of time, you must elicit interest in the problem, provide key context (what, why, how) and demonstrate your achievements/results

Evaluation – 20% of project grade

- Quality and clarity of presentation
- Effectiveness of Q&A period in defending the results and/or evaluating the business impact of the project

Past Project Topics

- Customer segmentation for a bank to help improve targeting of marketing campaigns
- Sentiment analysis of financial news
- Skin lesion classification for cancer detection
- Detecting natural disasters through X/Twitter posts
- Bank account fraud detection
- Worker's Compensation insurance claims forecasting
- Wildfire detection via satellite images

Note: Some of these topics are easier than others to produce a strong project